

# One Coalition, Two Electorates

*A short, code-free tour of what the data showed about Reform UK's voters*

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## The question

There's a comfortable shorthand in political commentary: "the Reform voter." I wanted to know whether that single label hides more than one kind of person.

The starting idea is **cross-pressure**—a classic in electoral sociology. A voter can hold attitudes that pull against their own vote. Ford and Goodwin describe this kind of base as "left behind": socially hardline, but economically dependent on the state—a claim others have since contested (Evans and Mellon) or reframed around identity and education (Sobolewska and Ford, in *Brexitland*). Rather than assert the tension exists, I wanted to test whether it actually shows up *inside* Reform's 2024 voters—using machine learning, and using several methods so they could check each other.

## Where the data made me change course

Three honest moments shaped the result, and I've left them visible rather than tidied away.

**A plan the data wouldn't support.** I'd intended to study LGB Reform voters and their own equality attitudes. But the survey's split-ballot design meant that of 336 Reform voters, only fifteen identified as non-heterosexual, and just two of those were asked the relevant questions. Two is not a finding. So I reframed the question from *identity-versus-vote* to *attitude-versus-attitude*—the stronger question anyway.

**A judgement call over a metric.** After using PCA to find the main attitudinal axes, I clustered the voters. The usual metric (the silhouette score) was highest at two clusters—but two clusters lumped almost all Reform voters together and hid exactly what I was looking for. I chose three, transparently trading a little statistical neatness for a result that answered the question. (As it happens, the two-cluster optimum is the two-electorates story; the third cluster is just a small left-libertarian fringe.)

**A method that disagreed with itself.** A first autoencoder seemed to confirm the clusters neatly—until I noticed its result leaned on a geometric quirk. A second one, built to remove the quirk, disagreed. Instead of quietly keeping the flattering number, I followed the disagreement, and it led to the most useful part of the analysis: pinning down exactly how much to trust the split.

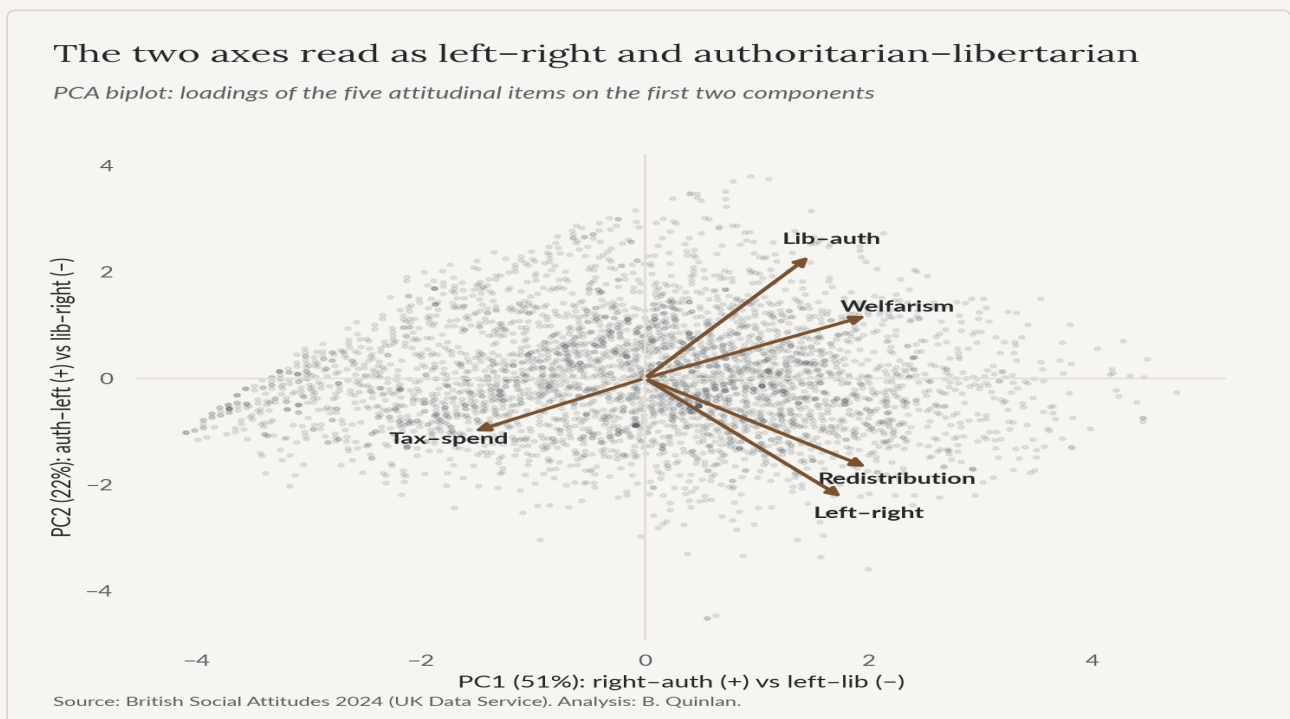
## The finding: two electorates, not one

Clustered on political attitudes alone, Reform's voters split almost in half:

- 55% sit in a conventional **right-authoritarian** group.

- **41%** sit in an **authoritarian-left** group—socially hardline, but economically left: strongly pro-redistribution and left-leaning, and with a majority receiving benefits.

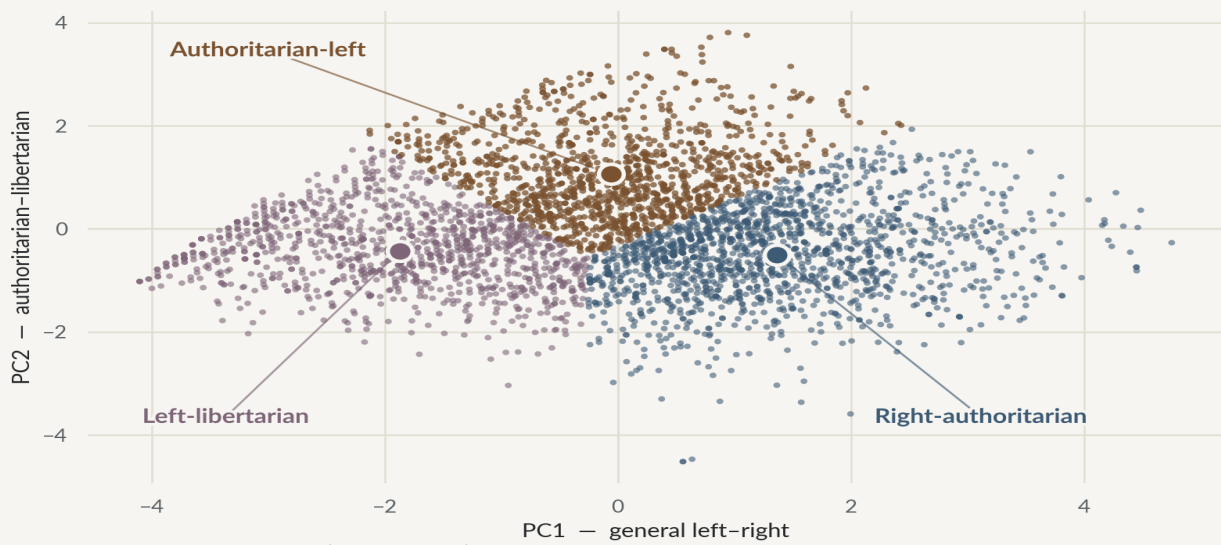
That second group is the cross-pressured one. Their economic instincts sit in real tension with the party they voted for—Reform campaigned in 2024 on roughly £90bn of tax cuts. But “economically left” undersells what’s actually going on. This group is pro-redistribution *and* about as sceptical of welfare claimants as the conventional right-authoritarian group *and* nearly as anti-immigration (around four in five, on the relevant split-ballot items, rate immigration bad for the economy). It’s their *benefit reliance*, not sympathy for claimants, that makes the “left-behind” label concrete. Redistribution to an implied in-group, alongside hostility to an out-group’s claim on the same welfare state, is what political scientists call **welfare chauvinism** (Andersen & Bjørklund, 1990)—a more precise description than “a left-wing electorate inside a right-wing party.”



The two attitudinal axes: PC1 (left–right fusion) and PC2 (authoritarian-left vs libertarian-right)

### Three attitudinal groups in Britain's 2024 electorate

*K-means clusters ( $k=3$ ) on the first two principal components. Reform's voters concentrate in two of them.*



The three-cluster solution: ~55% right-authoritarian, ~41% authoritarian-left

## Is it real—and is it old?

**How much to trust it.** The autoencoder that disagreed taught me where the uncertainty really lives. The split barely moves if I drop the weakest variable, cluster on the full data instead of a 2-D summary, or resample the respondents—a Reform voter placed in the left-behind group stays there across resamples 98% of the time. Hierarchical clustering agrees with the main method strongly, and a decision tree, coming at the problem backwards, finds the same dominant attitude and—tellingly—struggles to pick out the cross-pressured group, exactly as a genuinely conflicted group should. The one thing that *can* redraw the split is a

radically different way of compressing the data, and even then only at the soft border between the two Reform groups, never in whether the left-behind group exists.

**How old it is.** Projecting 2014 UKIP identifiers into the same model, they fall into the “left-behind” cluster *even more heavily* than 2024 Reform voters—59% against 41%, a gap well beyond chance. The pattern predates Reform by a decade. Looking closer, Reform’s coalition hasn’t *moved away* from the left-behind core so much as *added* a conventional-right layer on top of it: the left-behind group is still over-represented, while the conventional-right group flips from under- to over-represented over the ten years.

## What I’m claiming—and what I’m not

I’ve kept the claims honest. This is **descriptive**, the subgroups are small, and the clustering is unweighted—though reweighting the Reform split to the survey’s published weights barely changes it (52/44/4 against 55/41/4), so the two-electorates finding isn’t an artefact of that. These are not turnout predictions, and a repeated snapshot a decade apart can show the profile is durable without proving how individuals moved. But across linear and non-linear methods, supervised and unsupervised, and across ten years of data, the same structure keeps surfacing: a single “Reform voter” label hides two quite different electorates, and the cross-pressured half is real, robust, and at least a decade old.

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*Data: British Social Attitudes Survey 2024 (UK Data Service SN 9478) and 2014 (SN 7809), NatCen Social Research; English Indices of Deprivation 2019 (MHCLG); 2025 Local Elections Handbook (House of Commons Library). The raw survey microdata is licence-restricted and not included in this repository—see [data/README.md](#).*